Alkanes vs Alkenes

Functional Groups · Chemistry IGCSE

ALKANES

No Functional Group

Functional Group	❌ None — alkanes have no functional group
Bond Type	Only **C–C single bonds** and C–H bonds
Saturation	**Saturated** — all bonds are single bonds, no room for more
General Formula	C_nH_{2n+2}
Example	Ethane → CH_3-CH_3
Reactivity	Low — mainly combustion + substitution reactions

ALKENES

C=C Double Bond

Functional Group	✅ C=C — carbon-carbon double bond
Bond Type	Contains a **C=C double bond** — makes them reactive
Saturation	**Unsaturated** — the double bond can open up to add atoms

General Formula	C_nH_{2n}
Example	Ethene → $CH_2{=}CH_2$
Reactivity	**Higher** — undergoes addition reactions (Br_2, H_2, H_2O)

QUICK COMPARISON

Side by Side

PROPERTY	ALKANES	ALKENES
Functional Group	None	C=C
Saturated?	✓ Yes	✗ No
General Formula	C_nH_{2n+2}	C_nH_{2n}
Reactivity	Low	Higher
Bromine Water Test	Stays orange/brown	Decolorizes → clear

BROMINE WATER TEST

How to Tell Them Apart

Add bromine water (Br_2) to the compound:

+ Alkane

Stays **orange/brown**

No reaction — no double bond to react with

→

+ Alkene

Turns **clear/colorless**

Br_2 adds across the C=C double bond

KEY RULE

Double bond → Alkene → Has a functional group (C=C)
All single bonds → Alkane → No functional group